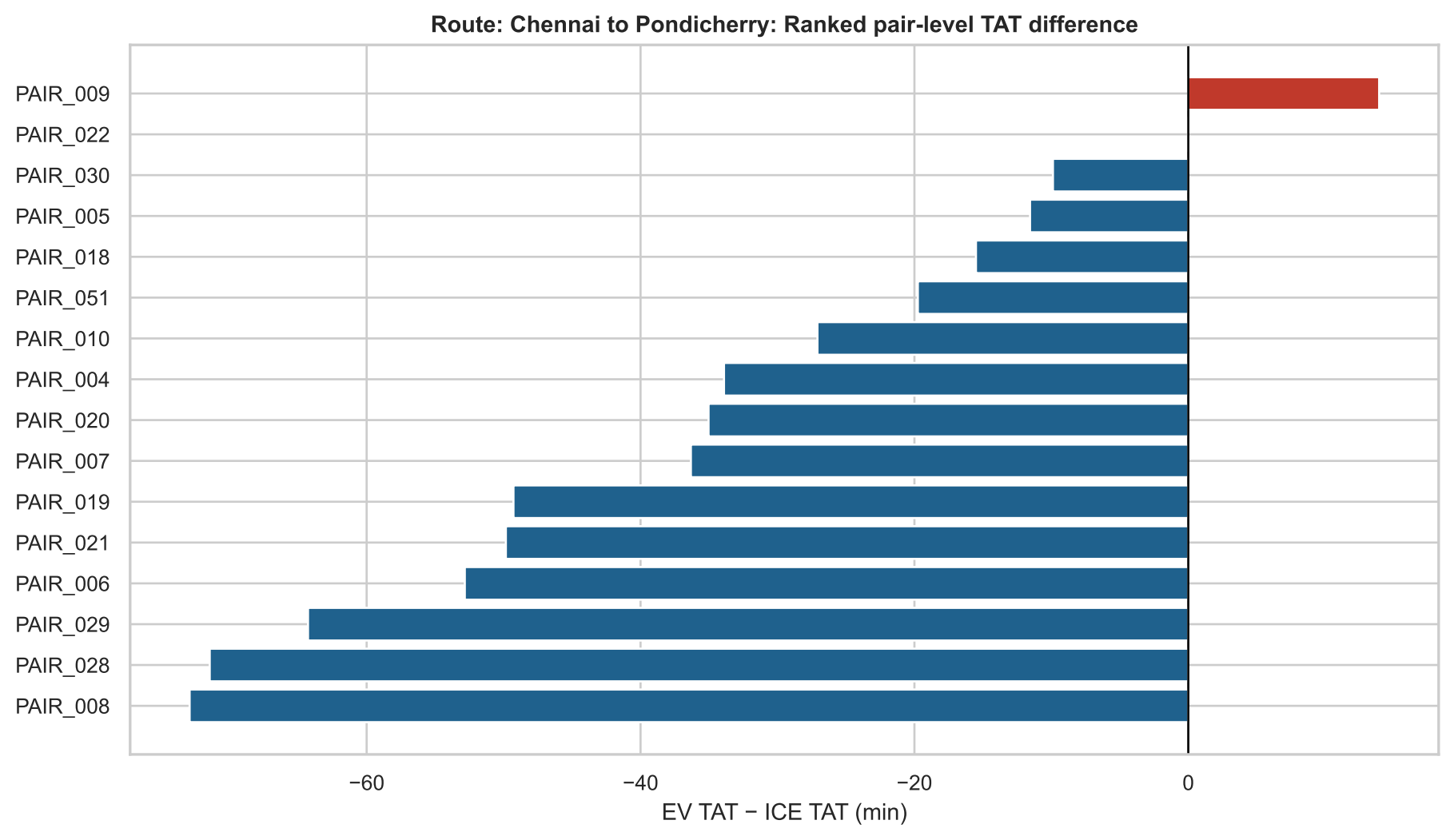
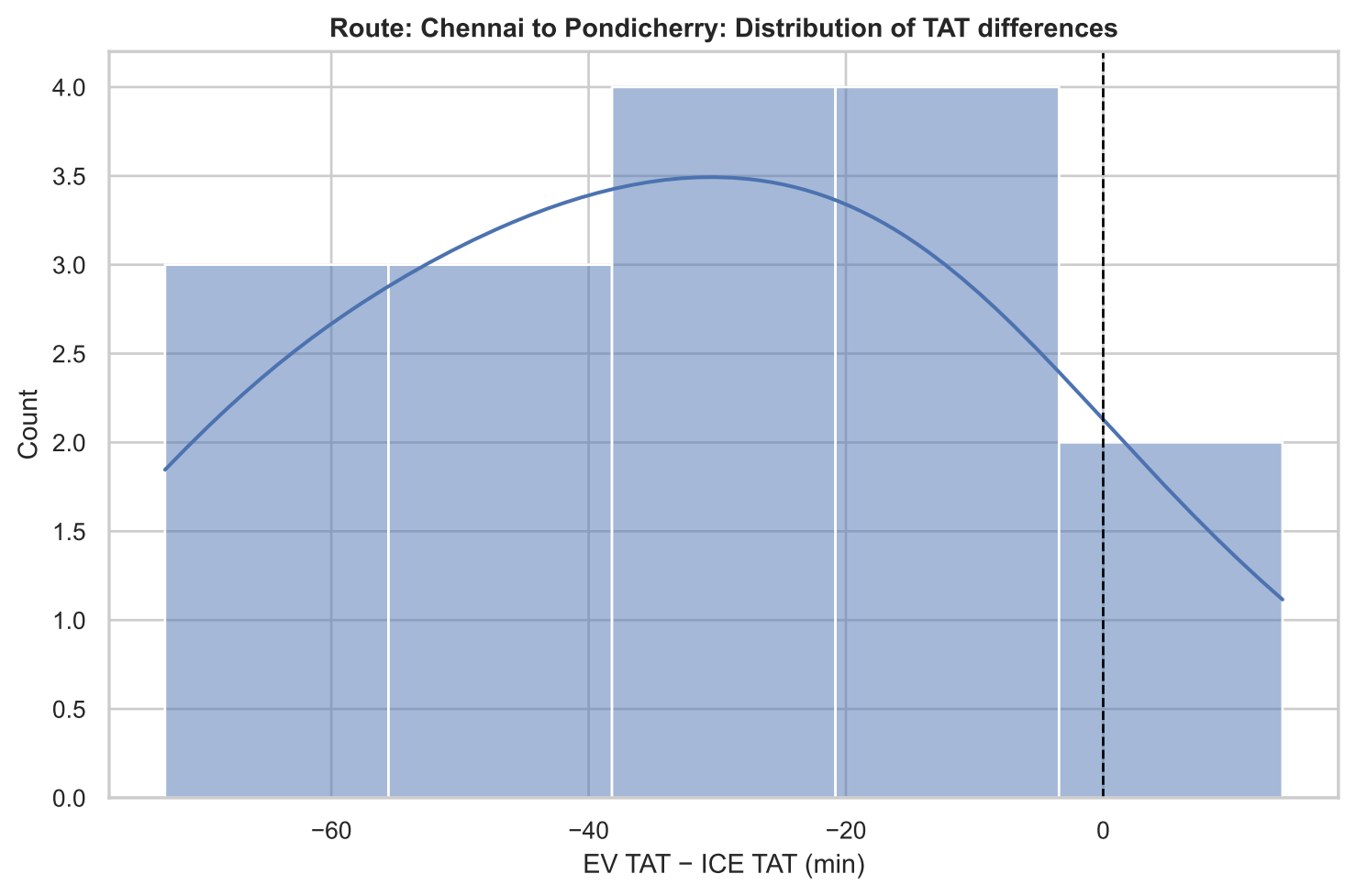
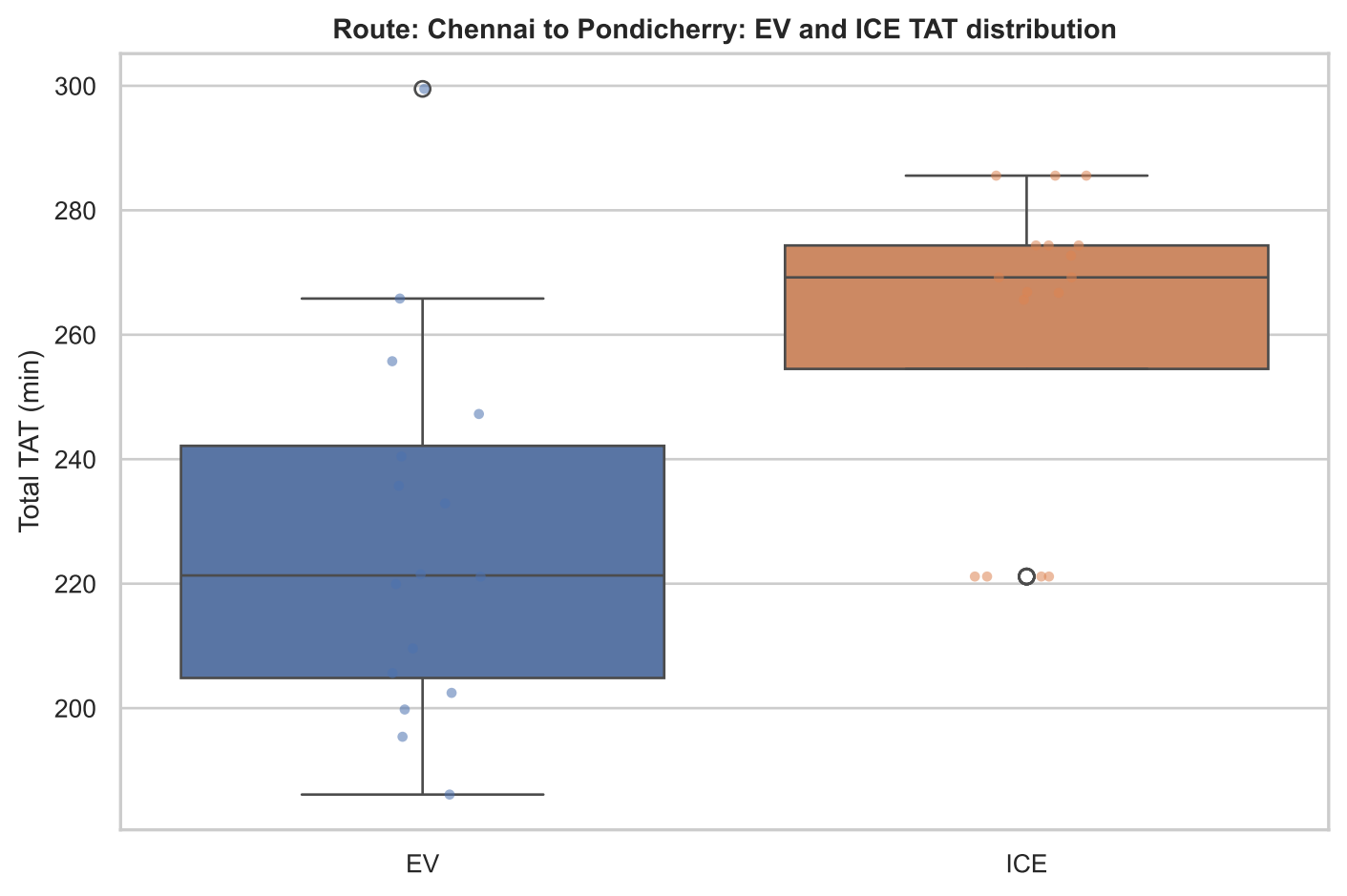
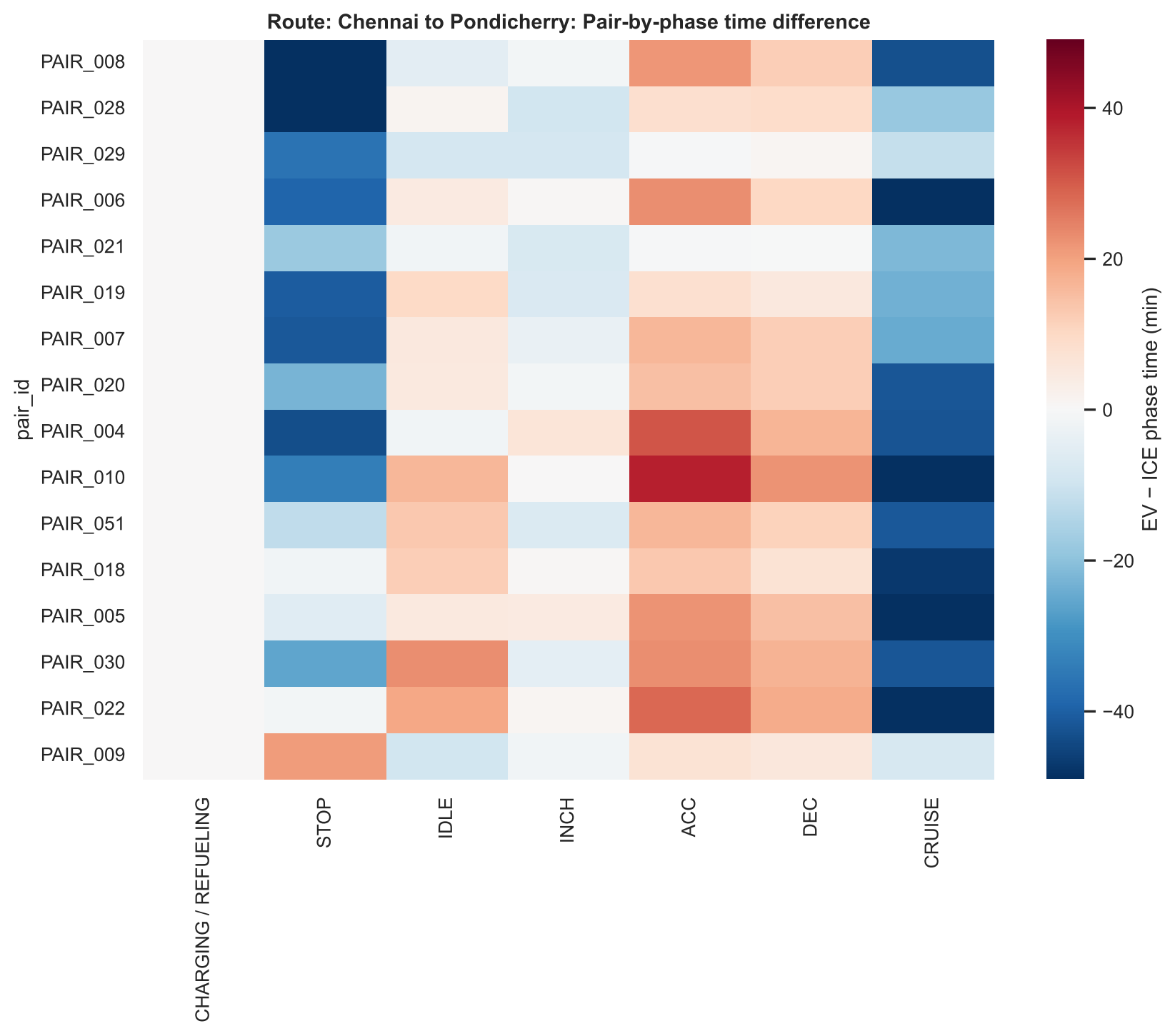
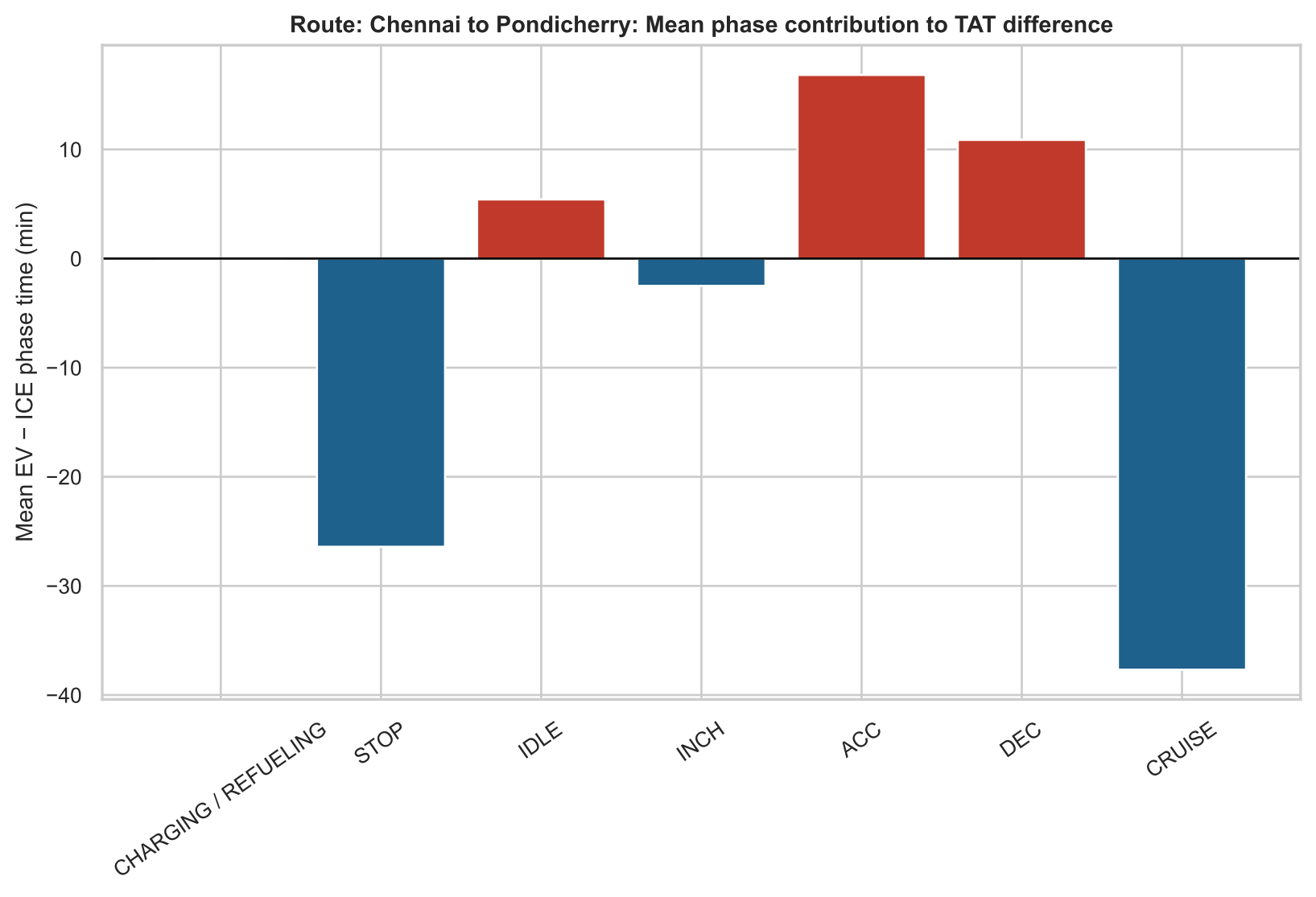
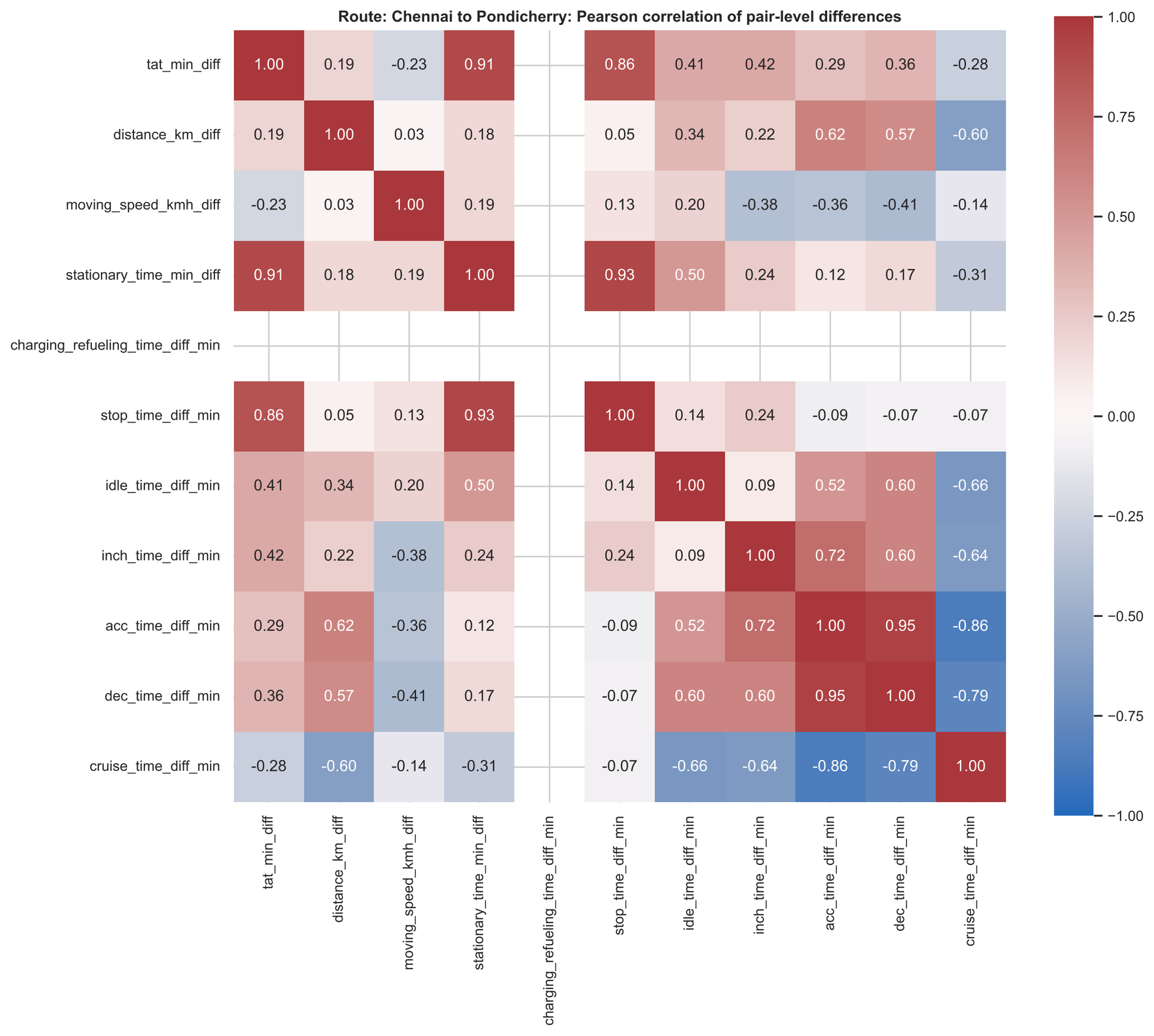

Route: Chennai to Pondicherry: Ranked pair-level TAT difference
PAIR_009
PAIR_022
PAIR_030
PAIR_005
PAIR_018
PAIR_051
PAIR_010
PAIR_004
PAIR_020
PAIR_007
PAIR_019
PAIR_021
PAIR_006
PAIR_029
PAIR_028
PAIR_008
−60
−40
−20
0
EV TAT − ICE TAT (min)

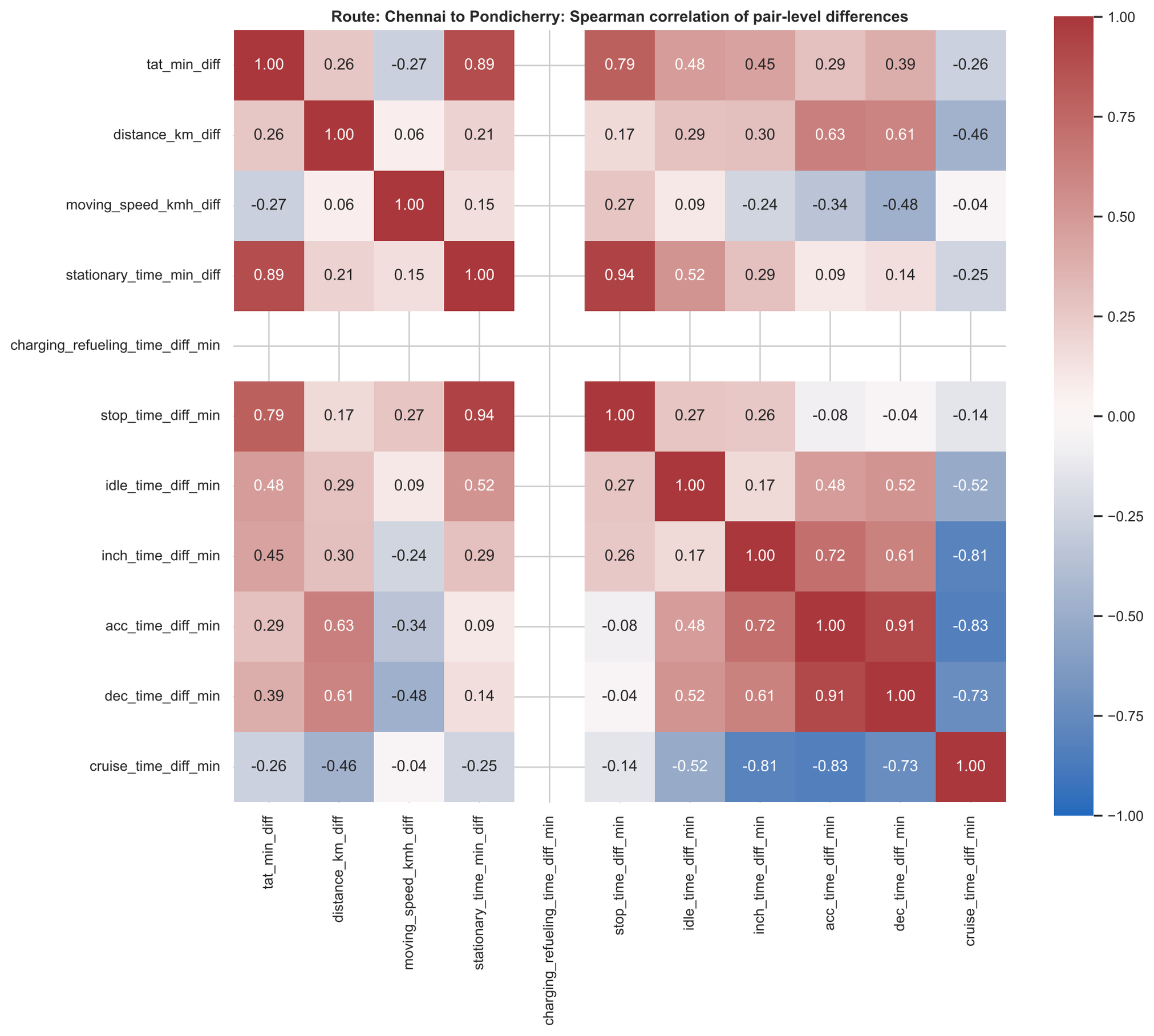

Route: Chennai to Pondicherry: Distribution of TAT differences
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−60
−40
−20
0
EV TAT − ICE TAT (min)

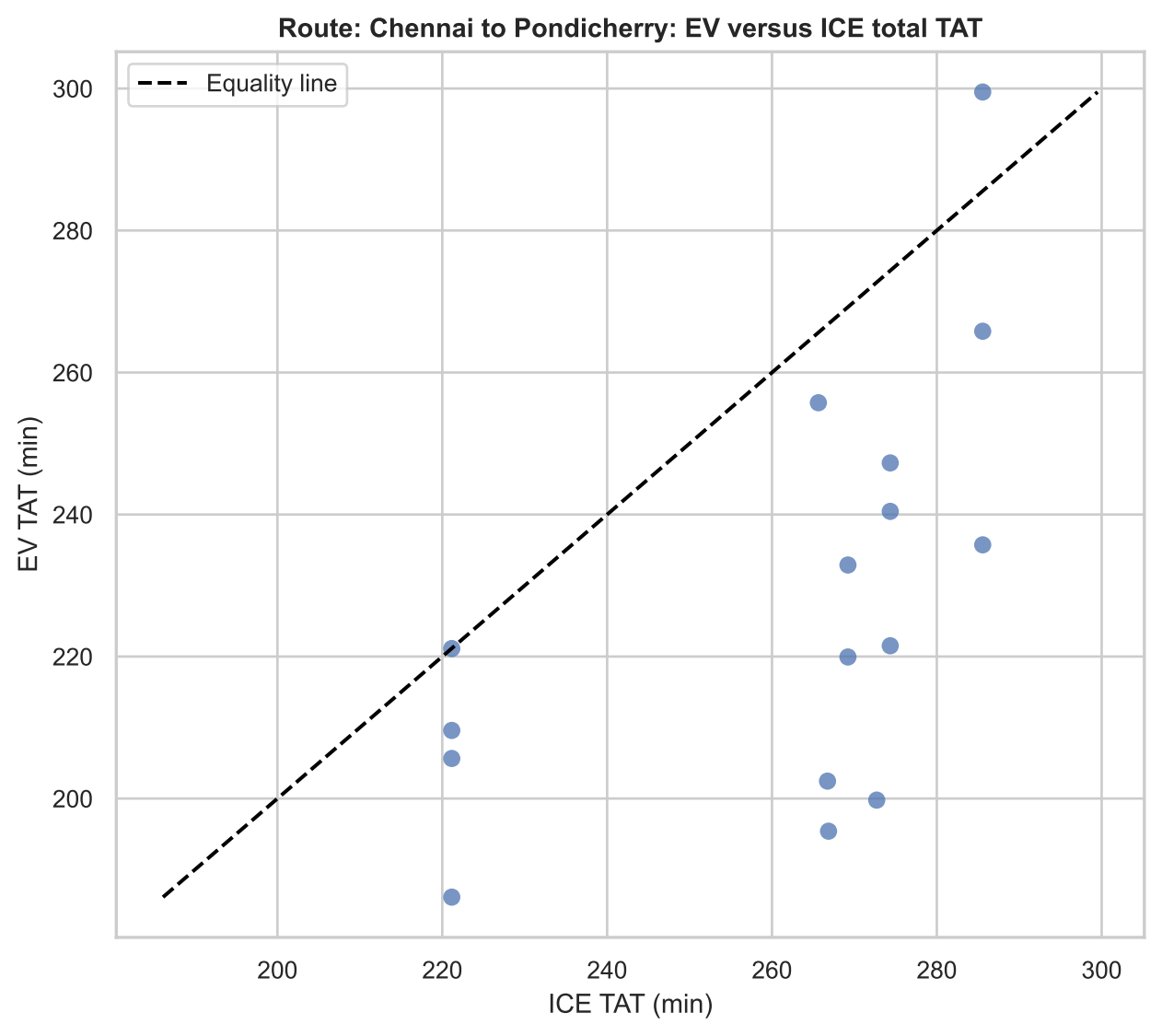

Route: Chennai to Pondicherry: EV and ICE TAT distribution
Total TAT (min)
300
280
260
240
220
200
EV
ICE

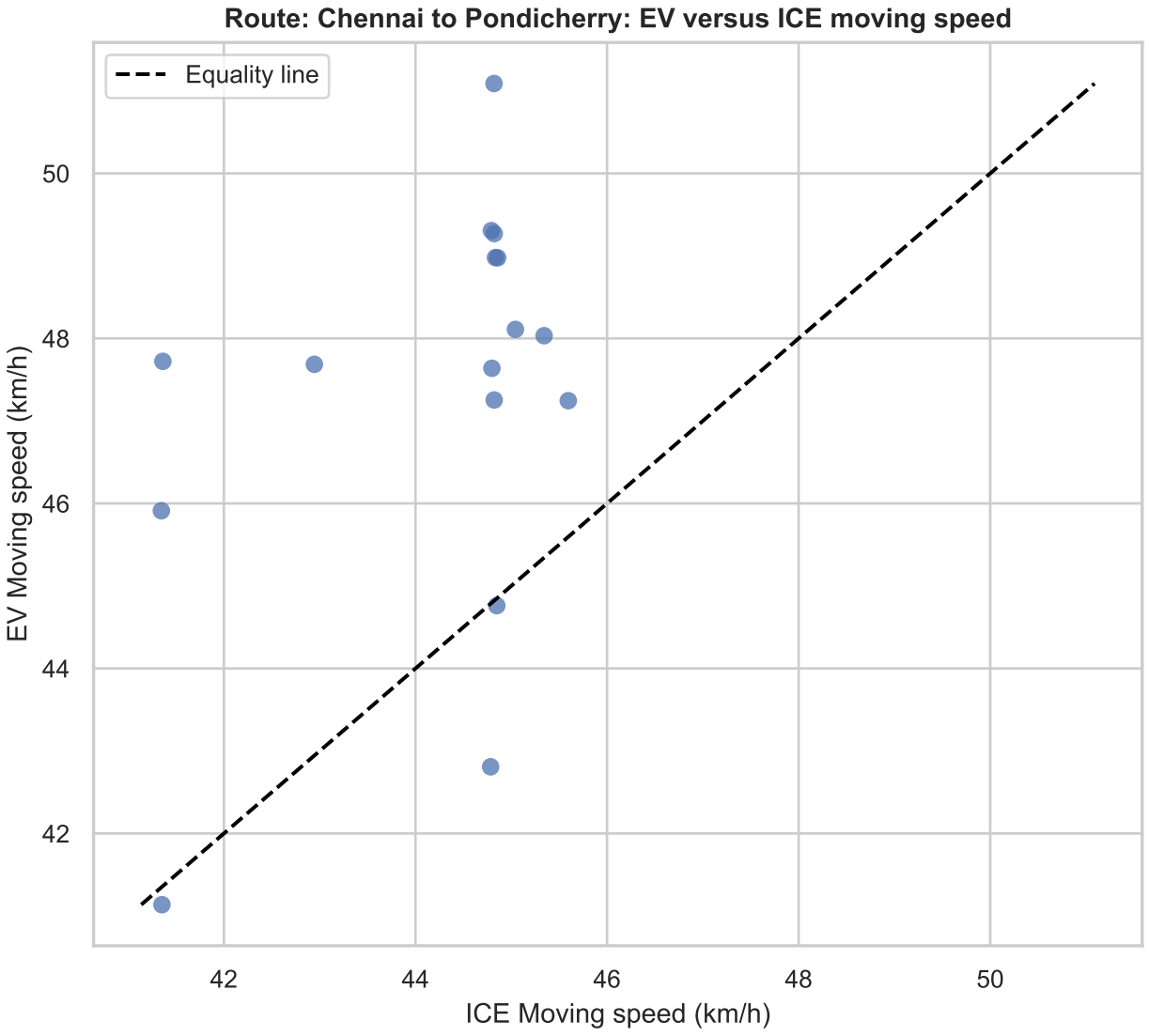

Route: Chennai to Pondicherry: Pair-by-phase time difference
pair_id
PAIR_008
PAIR_028
PAIR_029
PAIR_006
PAIR_021
PAIR_019
PAIR_007
PAIR_020
PAIR_004
PAIR_010
PAIR_051
PAIR_018
PAIR_005
PAIR_030
PAIR_022
PAIR_009
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

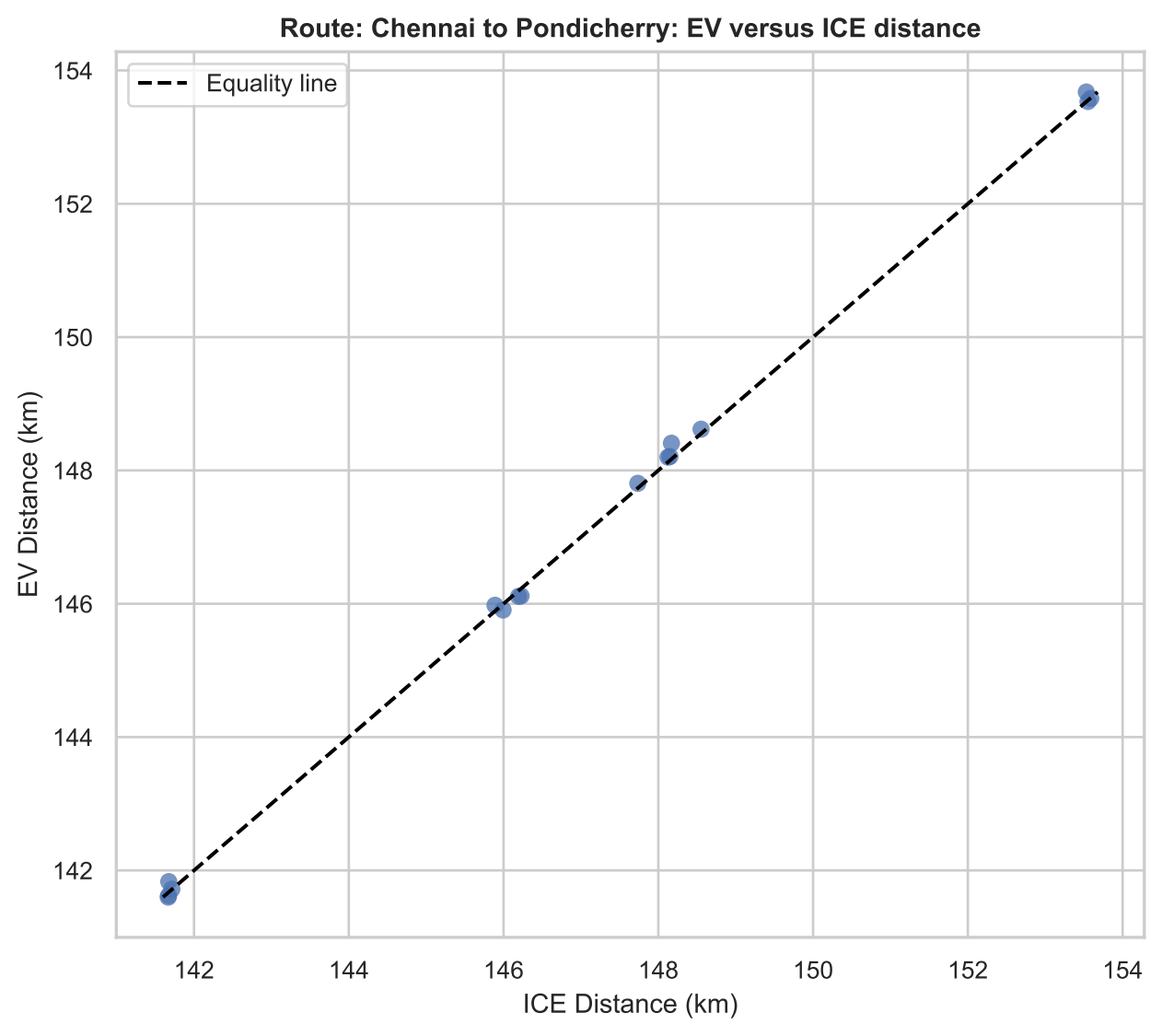

Route: Chennai to Pondicherry: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
10
0
−10
−20
−30
−40
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

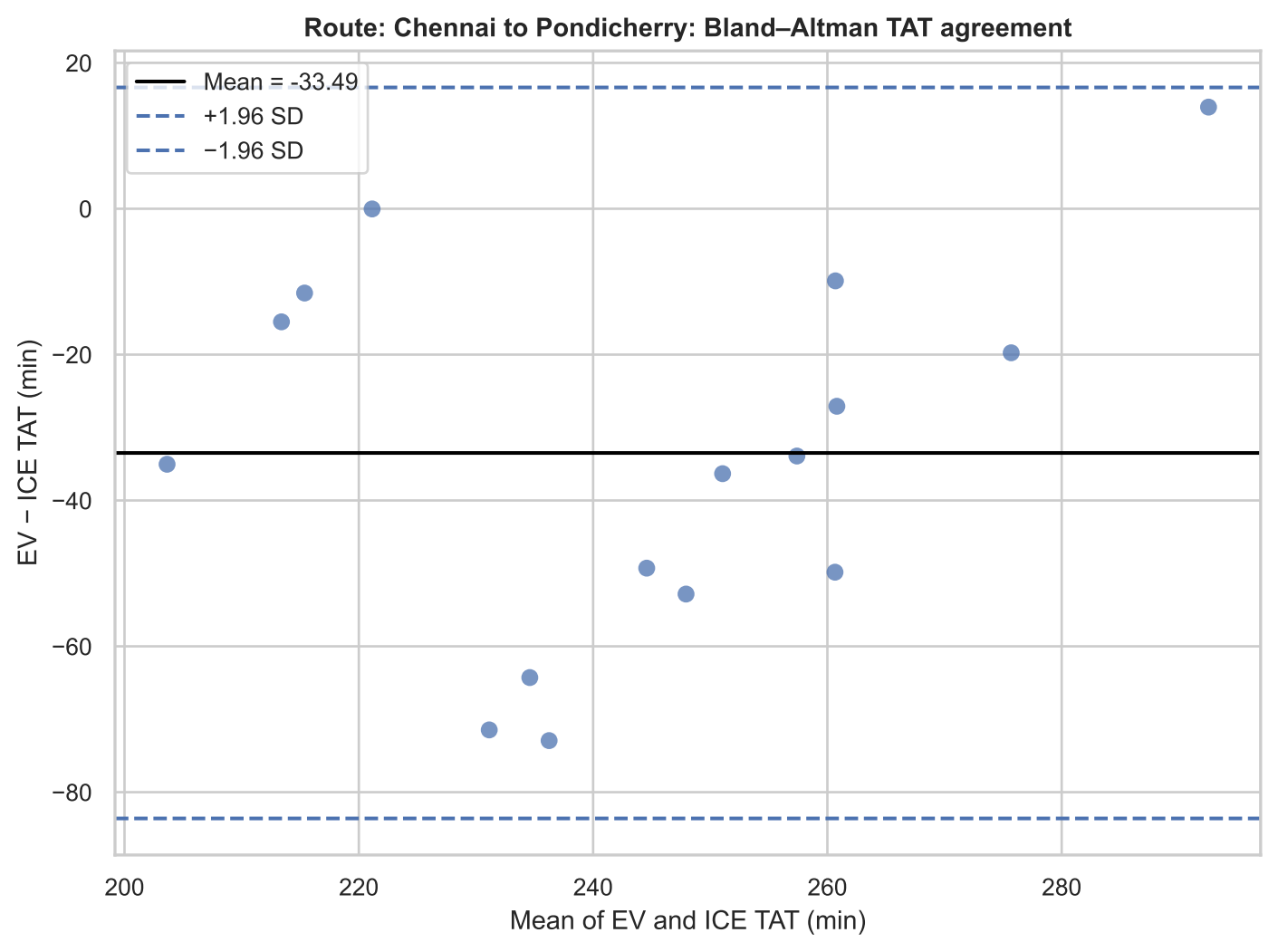

Route: Chennai to Pondicherry: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.19 -0.23 0.91 0.86 0.41 0.42 0.29 0.36 -0.28
0.19 1.00 0.03 0.18 0.05 0.34 0.22 0.62 0.57 -0.60
-0.23 0.03 1.00 0.19 0.13 0.20 -0.38 -0.36 -0.41 -0.14
0.91 0.18 0.19 1.00 0.93 0.50 0.24 0.12 0.17 -0.31
0.86 0.05 0.13 0.93 1.00 0.14 0.24 -0.09 -0.07 -0.07
0.41 0.34 0.20 0.50 0.14 1.00 0.09 0.52 0.60 -0.66
0.42 0.22 -0.38 0.24 0.24 0.09 1.00 0.72 0.60 -0.64
0.29 0.62 -0.36 0.12 -0.09 0.52 0.72 1.00 0.95 -0.86
0.36 0.57 -0.41 0.17 -0.07 0.60 0.60 0.95 1.00 -0.79
-0.28 -0.60 -0.14 -0.31 -0.07 -0.66 -0.64 -0.86 -0.79 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

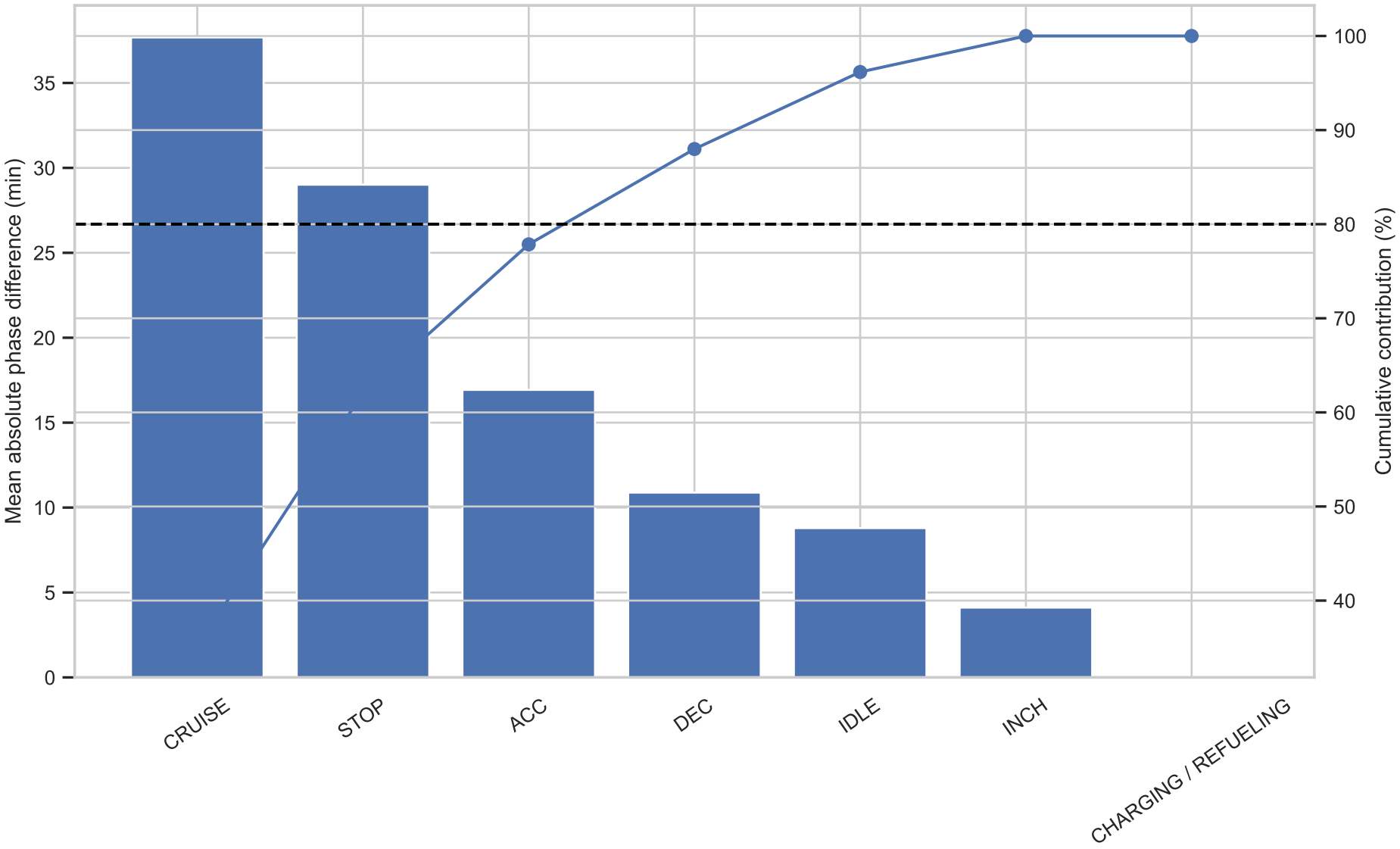

Route: Chennai to Pondicherry: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.26 -0.27 0.89 0.79 0.48 0.45 0.29 0.39 -0.26
0.26 1.00 0.06 0.21 0.17 0.29 0.30 0.63 0.61 -0.46
-0.27 0.06 1.00 0.15 0.27 0.09 -0.24 -0.34 -0.48 -0.04
0.89 0.21 0.15 1.00 0.94 0.52 0.29 0.09 0.14 -0.25
0.79 0.17 0.27 0.94 1.00 0.27 0.26 -0.08 -0.04 -0.14
0.48 0.29 0.09 0.52 0.27 1.00 0.17 0.48 0.52 -0.52
0.45 0.30 -0.24 0.29 0.26 0.17 1.00 0.72 0.61 -0.81
0.29 0.63 -0.34 0.09 -0.08 0.48 0.72 1.00 0.91 -0.83
0.39 0.61 -0.48 0.14 -0.04 0.52 0.61 0.91 1.00 -0.73
-0.26 -0.46 -0.04 -0.25 -0.14 -0.52 -0.81 -0.83 -0.73 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

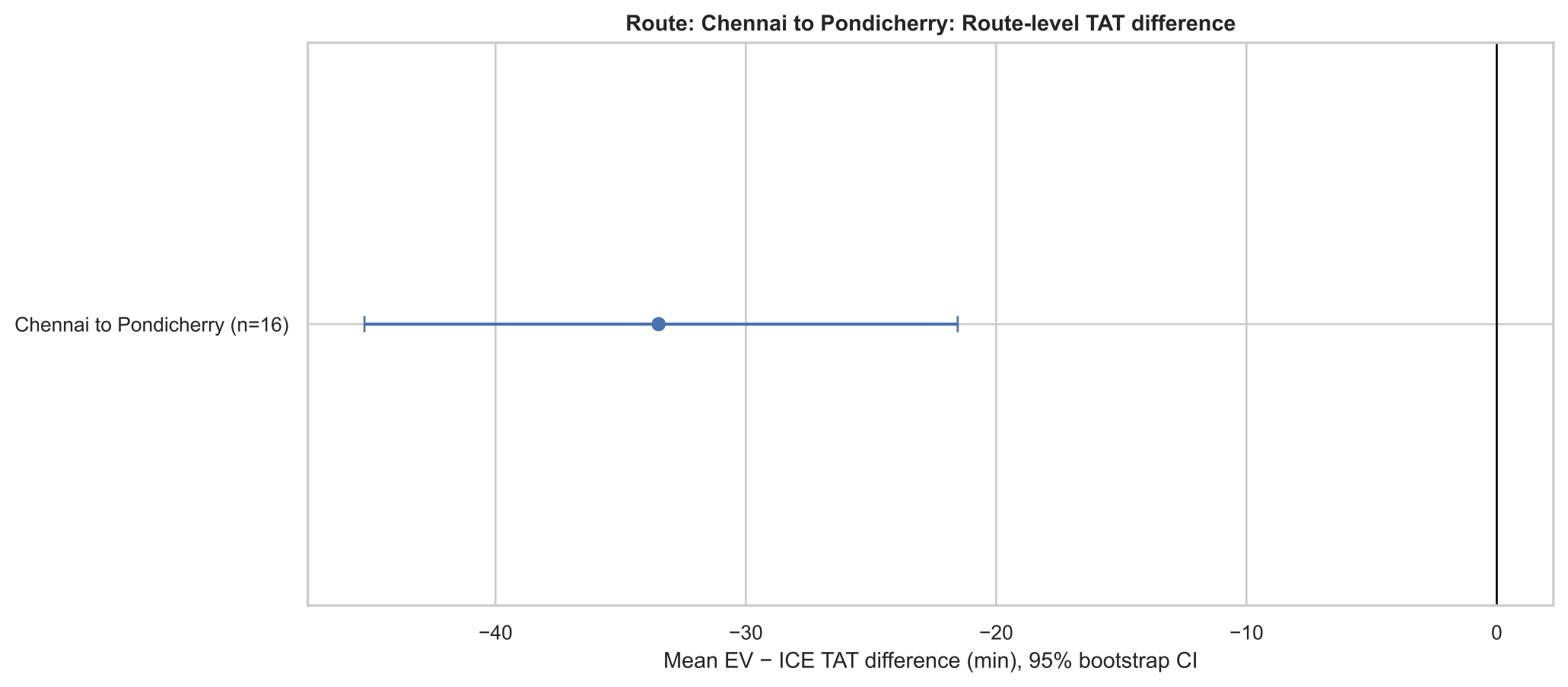

Route: Chennai to Pondicherry: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
300
280
260
240
220
200
200
220
240
260
280
300

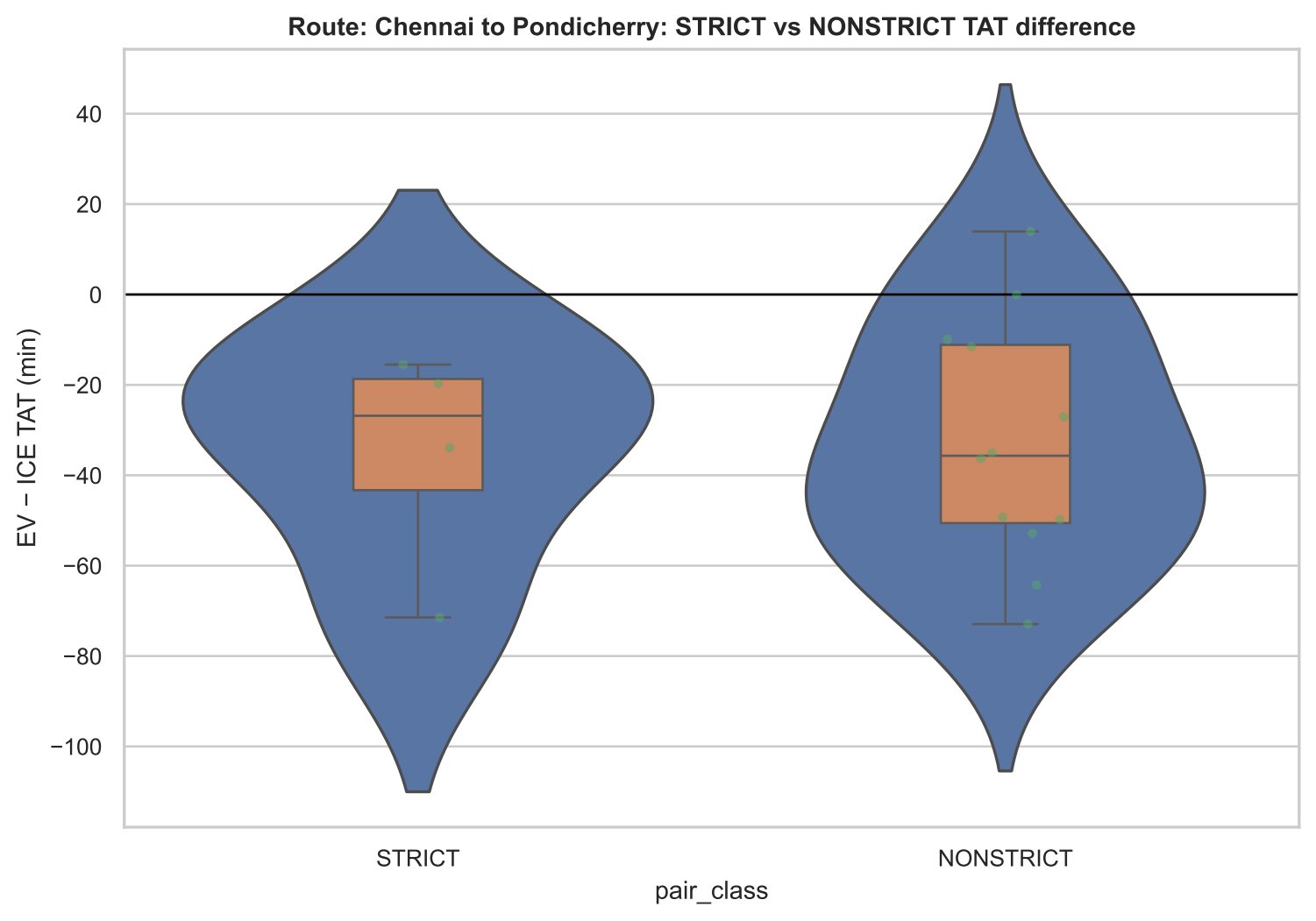

Route: Chennai to Pondicherry: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
42
44
46
48
50

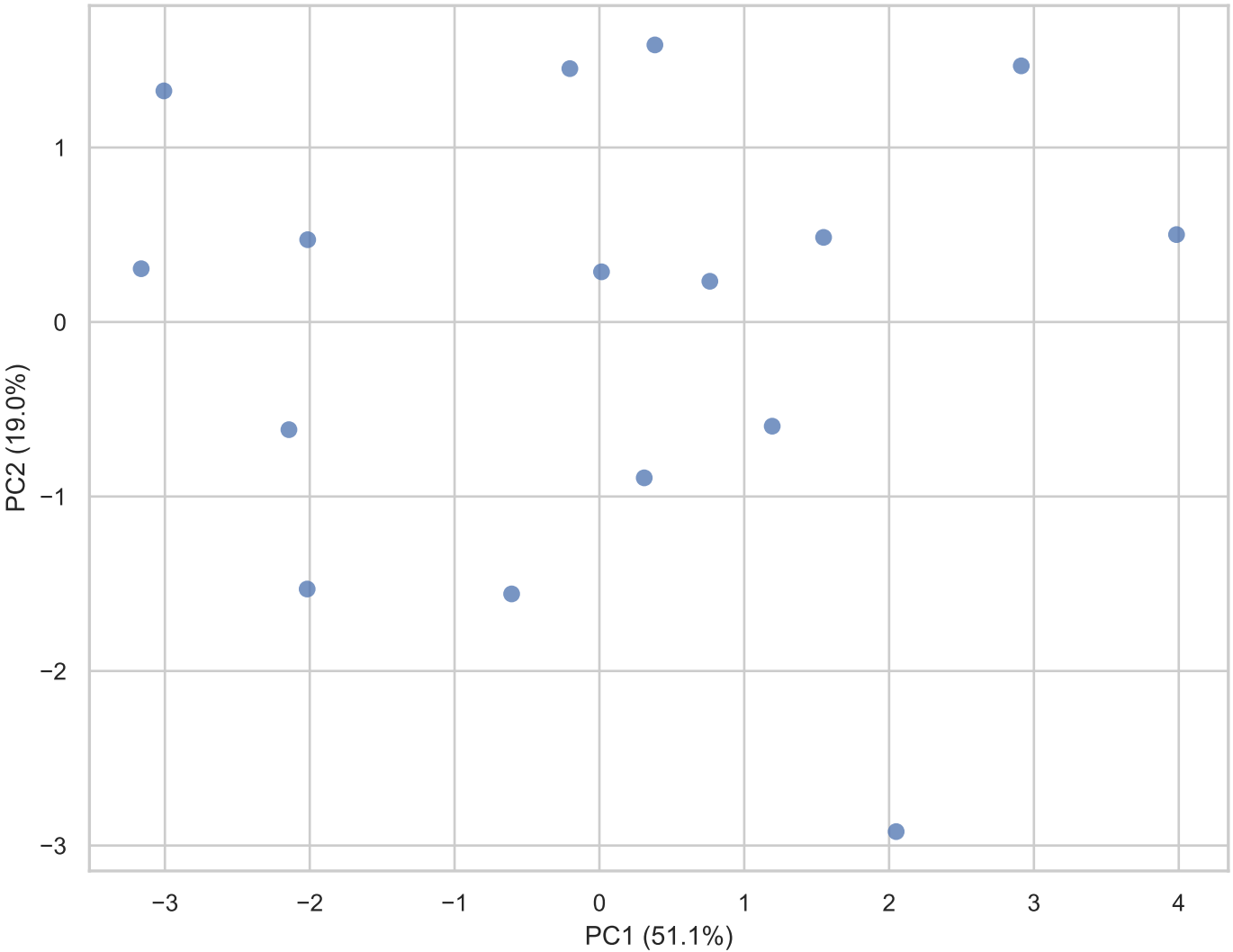

Route: Chennai to Pondicherry: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
154
152
150
148
146
144
142
142
144
146
148
150
152
154

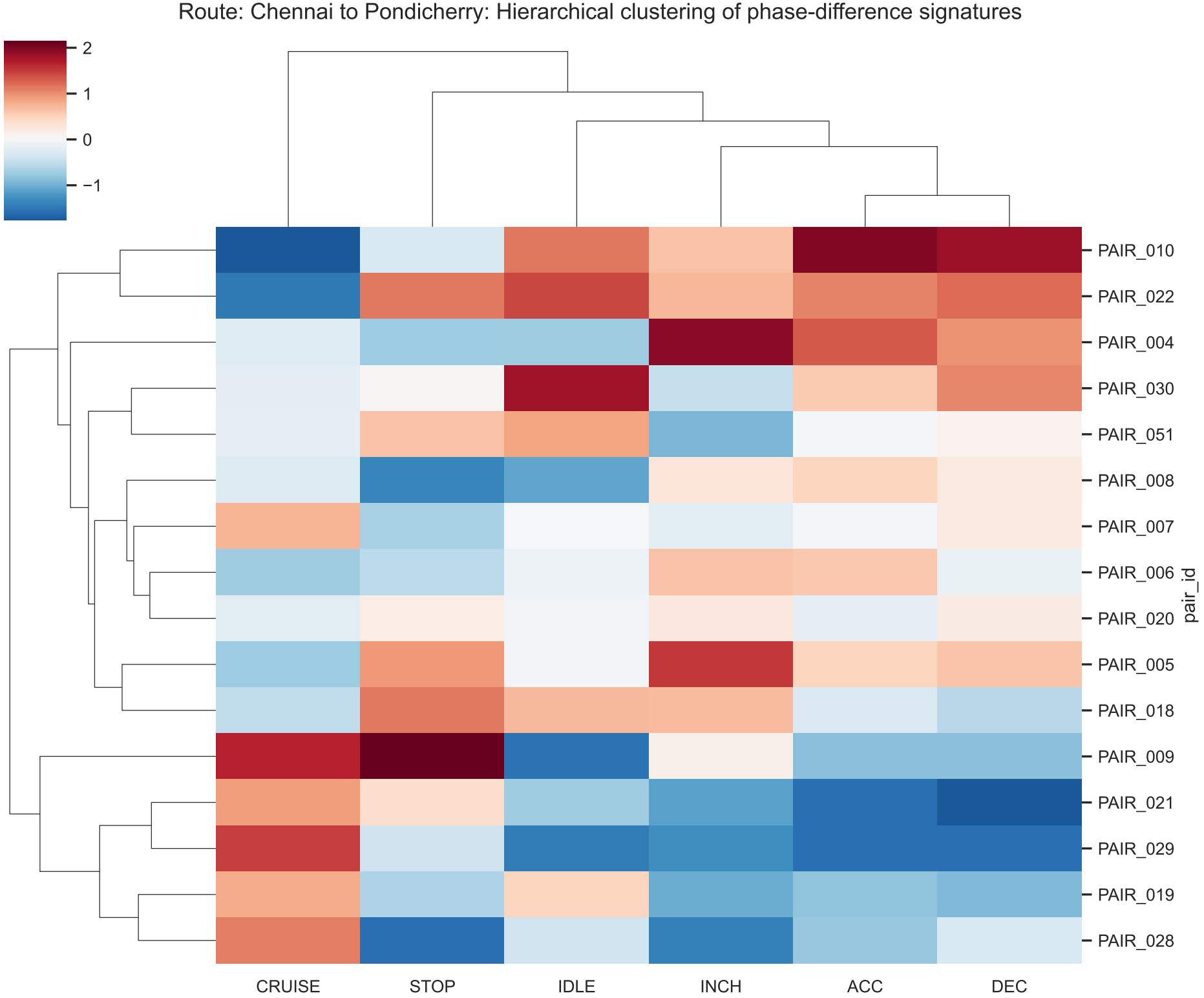

Route: Chennai to Pondicherry: Bland–Altman TAT agreement
Mean = -33.49
+1.96 SD
−1.96 SD
20
0
−20
−40
−60
−80
EV – ICE TAT (min)
200
220
240
260
280
Mean of EV and ICE TAT (min)


Route: Chennai to Pondicherry: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
35
40
50
60
70
80
90
100
CRUISE
STOP
ACC
DEC
IDLE
INCH
CHARGING / REFUELING


Route: Chennai to Pondicherry: Route-level TAT difference
Chennai to Pondicherry (n=16)
−40
−30
−20
−10
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Chennai to Pondicherry: STRICT vs NONSTRICT TAT difference
EV – ICE TAT (min)
40
20
0
−20
−40
−60
−80
−100
STRICT
NONSTRICT
pair_class


Route: Chennai to Pondicherry: PCA of EV−ICE operational differences
PC2 (19.0%)
PC1 (51.1%)
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


Route: Chennai to Pondicherry: Hierarchical clustering of phase-difference signatures
2
1
0
−1
PAIR_010
PAIR_022
PAIR_004
PAIR_030
PAIR_051
PAIR_008
PAIR_007
PAIR_006
PAIR_020
PAIR_005
PAIR_018
PAIR_009
PAIR_021
PAIR_029
PAIR_019
PAIR_028
pair_id
CRUISE
STOP
IDLE
INCH
ACC
DEC